

Differences Between System Analysis and System Design

DAY 136
23 / NOV / 2025

System Analysis and System Design are two stages of the software development life cycle. System Analysis is a process of collecting and analyzing the requirements of the system whereas System Design is a process of creating a design for the system to meet the requirements. Both are important stages as it helps to create an effective system with all the features and functions.

What is System Analysis?

System Analysis is a process of understanding the system requirements and its environment. It is one of the initial stages of the software development life cycle. System analysis is the process of breaking the system down into its individual components and understanding how each component interacts with the other components to accomplish the system's overall goal. In this process, the analyst collects the requirements of the system and documents them.

Characteristics

- It is the study of the existing system to identify the problem areas.
- It is a process of understanding the system requirements and its environment.
- It involves gathering and understanding the user's requirements.
- It involves analyzing the system in terms of its current and future needs.

Advantages

- It helps to identify the problems and their causes.
- It helps to understand the functional and non-functional requirements of the system.
- It helps to develop better solutions.
- It helps identify the areas of improvement.

Limitations

- It can be time-consuming.
- It can be costly.
- It can be difficult to get accurate information.

What is System Design?

System Design is the process of creating a design for the system to meet the requirements. System design is the process of designing the architecture, components, modules, interfaces, and data for a system to satisfy the specified requirements. It involves the design of the system architecture, components, modules, interfaces, and data.

Characteristics

- It is the process of creating a design for the system.
- It involves the design of the system architecture, components, modules, interfaces, and data.
- It involves identifying the modules and components of the system.
- It involves creating the user interface and database design.

Advantages

- It helps to create an efficient system.
- It helps identify the areas of improvement.
- It helps to reduce the development cost.
- It helps to create a better user experience.

Limitations

- It can be time-consuming.
- It can be costly.
- It can be difficult to get accurate information.

Differences between System Analysis and System Design

Factors	System Analysis	System Design
Purpose	System Analysis is the process of gathering and analyzing information to assess the suitability of a current system and to determine the requirements of a new system.	System Design is the process of specifying elements of a system such as modules, architecture, components, and their interfaces.
Approach	System Analysis is a top-down approach where the analyst looks at the big picture first and then delves into the details.	System Design is a bottom-up approach where the analyst starts with the details and moves up to the big picture.
Scope	System Analysis focuses on the needs of the user, the current system, and the business processes that the system must support.	System Design focuses on the design of the system, its architecture, and the components that make up the system.
Output	System Analysis produces the requirements document that describes the desired system.	System Design produces the design document that describes the architecture and components of the system.
Time	System Analysis is a one-time process that occurs at the beginning of the project.	System Design is an ongoing process that occurs throughout the project.
Methodology	System Analysis relies on a structured approach such as the Waterfall Model or the Agile Methodology.	System Design relies on an iterative approach such as the Spiral Model.
Tools	System Analysis utilizes tools such as interviews, surveys, questionnaires, and	System Design utilizes tools such as data flow diagrams and object-

	observation.	oriented diagrams.
Process	System Analysis is the first step in the software development process.	System Design is the second step in the software development process.
Goals	The goal of System Analysis is to identify and understand the user requirements and the business processes that the system must support.	The goal of System Design is to create a design that meets the user requirements and supports the business processes.
Risk	System Analysis involves minimal risk.	System Design involves significant risk, as the design may not meet the user requirements or support the business processes.
Problem Solving	System Analysis focuses on problem identification and definition.	System Design focuses on problem-solving and finding solutions.

Conclusion

System Analysis and System Design are two important stages of the software development life cycle. System Analysis is a process of collecting and analyzing the requirements of the system whereas System Design is a process of creating a design for the system to meet the requirements. System Analysis helps to identify the problems and their causes while System Design helps to create an efficient system. Both are necessary for creating an effective system with all the features and functions.